Ellian Carlos Oliveira Costa

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Experience

Software Development Engineer II **May 2024 - Present**
Amazon

- Solving end-to-end supply chain compliance problems with fully scalable systems that serves LATAM with 100M+ events a month and 1M+ invoices a month.
- Scaled the event-sourcing pipeline to 1M+ messages/day (100+ TPS), making its throughput and ordering guarantees the foundation other compliance services build on.
- Designed a Neptune graph projected from the event stream to model cross-entity relationships relational schemas couldn't express, enabling property-similarity matching across company-wide projects.
- Designed and shipped an MCP-backed first-responder AI agent for invoice-compliance tickets, cutting MTTA 70% and MTTR 13% while improving on-call transparency and customer communication.
- Diagnosed a silent data anomaly in the event pipeline, tracing a $1M+ cash-flow leak to its root cause and shipping the fix, recovering the cash.
- Owned operability for the domain (alarms, health/customer metrics, runbooks) and built org-wide onboarding: 15+ curated materials and a knowledge-transfer process that reduced bus factor across teams.
- Interviewer; conducted 20+ SWE interviews and graduated 2 shadow interviewers.

Senior Software Engineer, Backend **Dec 2022 - April 2024**
CI&T - Fintech Customer

- Designed event-driven, reactive microservices in AWS sustaining 99.9% availability and fault tolerance at 10M+ requests/day, serving 3M+ banking customers across Brazil.
- Built saga-pattern orchestration for multi-step customer journeys across 10+ microservices, handling partial-failure and compensation logic for financial operations that couldn't tolerate inconsistency.
- Designed a frugal account-lifecycle (deletion) orchestrator, saving 1M+ R$ over three years.
- Defined the DynamoDB data model (single-DB-per-capability, minimal replication) and hexagonal adapters reused across systems, establishing the team's reference architecture for new services.
- Drove technical direction as backend reference (TypeScript, design patterns, architecture review); mentored a junior engineer to mid-level in one year, while delivering onboarding journeys that reached 901k+ banking users.

Fullstack Developer **Mar 2020 - Dec 2022**
Tokenlab

- Designed and implemented multi-provider authentication (OAuth2 - Google/Facebook/Apple) for 1M+ users, raising conversion 8% and cutting password recoveries 12%.
- Optimized a 1TB+ SQL schema and query layer, improving GraphQL API latency up to 80% for curated mobile data.
- Built backend infrastructure for cross-device in-app purchases (sync, tax handling), increasing conversion over 20%.
- Built a backend-for-frontend with multiple microservices serving 10k+ clients/month across Web and Mobile.

Education

B.S. Information Systems - with Academic Honour **2020 - 2023**
Universidade de São Paulo (ICMC) · GPA 9.0/10.0

M.Sc. Computer Science - Software Engineering, Open Source **2026 - Present**
Universidade de São Paulo

Skills

Languages: Kotlin, TypeScript/JavaScript, Python, C/C++, SQL.
Distributed Systems: event sourcing, CQRS, saga and orchestration, high availability, system design, data modeling, observability.
AWS: DynamoDB, Lambda, SQS, SNS, CloudWatch, API Gateway, ECS/EKS, S3, Aurora, IAM.
Tooling: AWS CDK, CloudFormation, Docker, GitHub Actions, CI/CD, Redis, GraphQL, Nix, Testing).
AI: GenAI agents, MCPs, agent harness, agent security, prompting, claude code.

Languages & Awards

Languages: Portuguese (native), English (fluent, C1), Spanish (intermediate), Japanese & Libras (beginner).
Awards: Academic Honour, USP (2024); Bronze - OBMEP 2018/2019, OBA 2018; Hon. Mention - OBMEP 2016/2017.